

# Customer Value Intelligence: A Data-Driven Retention Strategy

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## 1. Introduction

In today's highly competitive retail environment, customer retention has become as important as customer acquisition. While promotional campaigns and discounts are effective in driving short-term sales, excessive dependence on discounts can significantly reduce profit margins and may fail to build genuine customer loyalty. Consequently, businesses increasingly seek to identify customers who generate sustainable long-term value and distinguish them from customers who purchase primarily because of promotional incentives.

This project, "**Customer Value Intelligence: A Data-Driven Retention Strategy**," focuses on analyzing customer purchasing behaviour to understand the drivers of customer value and loyalty. Using customer demographic information, purchasing behaviour, promotional activity, and transaction characteristics, the project develops analytical metrics that enable data-driven customer segmentation and strategic business recommendations.

Rather than relying solely on descriptive statistics, the project combines **data cleaning, feature engineering, SQL-based business analysis, and Power BI visualization** to transform raw customer data into actionable business intelligence. The ultimate objective is to provide recommendations that improve customer retention while protecting business profitability.

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## 2. Problem Statement

Retail businesses frequently use promotional discounts to increase customer acquisition and sales volume. However, indiscriminate discounting may encourage transactional purchasing behaviour instead of genuine customer loyalty, leading to reduced margins and inefficient marketing expenditure.

The dataset used in this project contains customer demographic information, purchase behaviour, promotional activity, and transaction details. Although it provides valuable transactional information, it does not explicitly identify customer loyalty, customer lifetime value, promotion dependency, or ideal customer profiles.

The primary challenge of this project is therefore to construct meaningful business metrics from the available data and answer the following business questions:

- What distinguishes high-value customers from low-value customers?
- Which behavioural patterns indicate long-term customer value?
- Which customers are genuinely loyal and which purchase primarily because of discounts?

- Which customer segments should gradually receive fewer promotional incentives?
- What characteristics define the brand's ideal customer profile?

Addressing these questions requires not only descriptive analysis but also the development of business-oriented features capable of representing concepts that are not directly available within the raw dataset.

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## 3. Project Objectives

The primary objectives of this project are:

- Perform comprehensive data cleaning to improve data quality and consistency.
  - Engineer meaningful business features that capture customer value, loyalty, promotional dependency, and engagement.
  - Develop SQL-based analytical queries to answer strategic business questions.
  - Identify behavioural and demographic patterns associated with high-value customers.
  - Design an interactive Power BI dashboard to communicate business insights effectively.
  - Recommend actionable customer retention strategies and promotional policies that balance revenue growth with profitability.
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## 4. Methodology

The project follows a structured analytical workflow consisting of five major stages.

1. **Data Cleaning:** Identification and treatment of missing values, redundant attributes, inconsistent categorical values, and validation of data quality.
  2. **Feature Engineering:** Construction of business-oriented metrics such as Promotion Persona, Promotion Dependency Score, Value Tier, Loyalty Score, and demographic indicators to represent concepts not explicitly present in the dataset.
  3. **SQL-Based Analysis:** Development of structured SQL queries to answer the business questions related to customer value, purchasing behaviour, geographical opportunity, and promotional effectiveness.
  4. **Dashboard Development:** Visualization of analytical findings using Microsoft Power BI through customer segmentation, customer pyramid analysis, geographic opportunity mapping, loyalty analysis, and behavioural insights.
  5. **Business Recommendations:** Translation of analytical findings into actionable retention strategies, promotional sunset plans, and an ideal customer profile for future customer acquisition.
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## 5. Data Cleaning

Initially, the dataset had 3900 records and 18 columns

The following were the columns present in the dataset:

['Customer ID',  
'Age',  
'Gender',  
'Item Purchased',  
'Category',  
'Purchase Amount (USD)',  
'Location',  
'Size',  
'Color',  
'Season',  
'Review Rating',  
'Subscription Status',  
'Shipping Type',  
'Discount Applied',  
'Promo Code Used',  
'Previous Purchases',  
'Payment Method',  
'Frequency of Purchases']

The data cleaning phase focused on ensuring that the dataset was accurate, consistent, and suitable for subsequent feature engineering and analytical modeling. The raw dataset contained **3,900 customer records and 18 attributes**, with each **Customer ID** being unique, confirming that every row represented a distinct customer and that no duplicate records were present.

A detailed data quality assessment was performed to identify missing values, redundant information, and structural inconsistencies. The only missing values were found in the **Review Rating** column, where **37 records (0.95% of the dataset)** were missing. Since the proportion of missing data was very small and no systematic pattern was observed, **median imputation** was used to preserve all customer records while minimizing the effect of outliers.

Redundancy analysis revealed that **Promo Code Used** and **Discount Applied** contained identical information for every customer. To eliminate duplicate information and avoid double-counting during feature engineering, the redundant **Promo Code Used** column was removed. Cross-analysis between **Subscription Status** and **Discount Applied** further showed that all subscribers received discounts, while some non-subscribers also benefited from promotional offers. This relationship provided valuable business insight and later formed the basis for customer segmentation.

The categorical values of **Frequency of Purchases** were also standardized by consolidating equivalent labels such as **Fortnightly/Bi-Weekly** and **Quarterly/Every 3 Months**, ensuring consistency before numerical transformation. Additional validation checks confirmed that purchase amounts and previous purchases were within expected ranges, with every customer having at least one previous purchase, indicating that the dataset represented existing customers rather than first-time buyers. Finally, binary categorical variables such as **Subscription Status** and **Discount Applied** were converted into numerical format to support SQL analysis and feature engineering.

Overall, these preprocessing steps improved data quality, removed redundancy, standardized categorical information, and prepared the dataset for reliable feature engineering, customer segmentation, SQL querying, and Power BI visualization.

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## 6. Feature Engineering

### 6.1 Overview

The raw dataset provided customer demographic information, purchasing behavior, promotional activity, and transaction details. However, the business objectives of the project required insights into concepts such as **customer value, loyalty, promotion dependency, and customer segmentation**, none of which were explicitly available in the dataset. Therefore, a feature engineering phase was performed to transform the raw variables into meaningful business metrics that could support SQL-based analysis, customer segmentation, and strategic decision-making.

The engineered features were designed to satisfy the project's key analytical objectives, including identifying high-value customers, distinguishing genuine loyalty from promotion-driven purchases, evaluating customer engagement, and supporting the development of an actionable promotional strategy.

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### 6.2 Promotion Persona

A new categorical feature, **promo\_persona**, was created by combining **Subscription Status** and **Discount Applied**.

Three customer groups were identified:

- **Subscriber** – Customers enrolled in the subscription program who received discounts.
- **Promo Opportunist** – Non-subscribers who purchased only with promotional discounts.
- **Organic** – Customers who purchased without any promotional incentive.

This feature enables the differentiation between customers who are organically loyal and those whose purchases are primarily driven by promotions. It forms the foundation for analyzing promotion effectiveness and evaluating customer dependence on discounts.

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### 6.3 Purchase Frequency Score

The original **Frequency of Purchases** attribute consisted of categorical values such as *Weekly*, *Monthly*, and *Annually*, which are unsuitable for quantitative analysis.

Each category was therefore converted into its annual purchase equivalent:

| Frequency | Score |
|-----------|-------|
| Weekly    | 52    |
| Bi-Weekly | 26    |
| Monthly   | 12    |
| Quarterly | 4     |
| Annually  | 1     |

This transformation preserves the true magnitude of purchasing behavior instead of assigning arbitrary ordinal rankings. The resulting feature provides a numerical representation of customer engagement and enables meaningful comparison across customers.

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## 6.4 Tenure Tier

Although **Previous Purchases** indicates customer experience with the brand, interpreting individual numerical values is difficult from a business perspective.

Customers were therefore grouped into four lifecycle stages:

- New
- Growing
- Established
- Loyal

The segmentation was based on purchase-count quartiles to ensure balanced customer distribution across groups.

This feature facilitates lifecycle analysis and enables comparison of purchasing behavior across different stages of customer maturity.

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## 6.5 Promotion Dependency Score

One of the primary objectives of the project was to determine which customers genuinely exhibit brand loyalty and which continue purchasing primarily because of promotional incentives.

Since no direct measure of promotion dependency existed within the dataset, a composite **Promotion Dependency Score** was constructed.

The score combines:

- Discount Applied

- Purchase Frequency Score
- Previous Purchases

Purchase frequency and previous purchases were first normalized using Min-Max normalization to ensure equal contribution despite differing scales.

The dependency score was computed as:

$$[ \text{Promotion} \backslash \text{Dependency} = (\text{Discount}) \backslash \text{times} \backslash \text{left} (1 - 0.5 \backslash \text{times} \text{Frequency\_}\{ \text{norm} \} - 0.5 \backslash \text{times} \text{Tenure\_}\{ \text{norm} \} \backslash \text{right} ) ]$$

The resulting score ranges from **0 to 1**, where:

- **0** represents customers with little or no dependence on promotional incentives.
- **1** represents customers exhibiting the highest estimated dependence on discounts.

Customers who never received discounts automatically obtain a dependency score of zero, while customers receiving discounts but demonstrating weak purchasing history and low engagement receive higher dependency values.

This feature directly supports the project's objective of designing a promotional sunset strategy.

## 6.6 Value Tier

The dataset did not contain any direct measure of customer value or Customer Lifetime Value (CLV). Consequently, a proxy value metric was developed by combining three behavioral dimensions:

- Purchase Amount
- Previous Purchases
- Purchase Frequency Score

The composite value metric was calculated as:

$$[ \text{Raw} \backslash \text{Value} = \text{Purchase} \backslash \text{Amount} \backslash \text{times} \text{Previous} \backslash \text{Purchases} \backslash \text{times} \text{Purchase} \backslash \text{Frequency} \backslash \text{Score} ]$$

Rather than using arbitrary thresholds, customers were ranked according to percentile and divided into four equally sized value segments:

- Low
- Medium
- High
- Premium

This approach identifies customers contributing the greatest long-term commercial value rather than simply those making the largest individual purchases. The feature serves as the basis for customer pyramid analysis and the identification of the brand's highest-value customer segments.

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## 6.7 Satisfaction Flag

Customer satisfaction was derived from the **Review Rating** attribute.

Ratings greater than or equal to **4.0** were classified as satisfied customers, while lower ratings were classified as non-satisfied.

This binary feature provides an easily interpretable measure of customer satisfaction and enables analysis of whether highly loyal customers are genuinely satisfied with the brand or primarily retained through promotional incentives.

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## 6.8 Age Group

The continuous **Age** variable was grouped into four demographic segments:

- 18–25
- 26–35
- 36–50
- 51–70

Age grouping simplifies demographic analysis and supports the development of an Ideal Customer Profile by identifying the most commercially valuable customer age segments.

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## 6.9 Premium Shipping Indicator

Shipping preferences were converted into a binary indicator identifying customers who selected premium shipping methods.

Customers choosing:

- Next Day Air
- Express
- 2-Day Shipping

were classified as premium shipping users.

The feature acts as a behavioral indicator of lower price sensitivity and supports the analysis of customer purchasing preferences.

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## 6.10 Digital Payment Indicator

Payment methods were grouped into two categories:

- Digital Payment
- Cash Payment

Digital payment users are generally more accessible through online marketing channels and loyalty programs. This feature assists in evaluating customer acquisition and retention strategies through digital engagement.

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## 6.11 Loyalty Score

Customer loyalty was not explicitly available within the dataset and therefore had to be approximated using behavioral indicators.

A composite **Loyalty Score** was developed using four components:

- Purchase Frequency
- Previous Purchases
- Promotion Dependency
- Customer Satisfaction

The score was calculated as:

$$\begin{aligned} &[ \\ &\text{Loyalty=} \\ &\text{Frequency}_{\{\text{norm}\}} \\ &\backslash \text{times} \\ &\text{Tenure}_{\{\text{norm}\}} \\ &\backslash \text{times} \\ &(1 - \text{Promotion} \backslash \text{Dependency}) \\ &\backslash \text{times} \\ &(1 + 0.2 \backslash \text{times Satisfaction}) \\ &] \end{aligned}$$

This formulation rewards customers who purchase frequently, maintain long-term relationships with the brand, demonstrate low dependence on promotions, and report higher satisfaction.



Unlike traditional loyalty measures based solely on purchase frequency, the proposed metric incorporates promotional behaviour and customer experience, providing a more comprehensive estimate of genuine customer loyalty.

The Loyalty Score became the principal metric used throughout the SQL analysis, Power BI dashboard, customer segmentation, and business recommendation phases.

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## 6.12 Summary

The feature engineering phase successfully transformed the cleaned transactional dataset into a business-oriented analytical dataset by constructing features that capture customer value, loyalty, promotional dependency, engagement, and demographic characteristics. Since these business concepts were not explicitly available in the original data, they were derived using meaningful combinations of existing customer attributes and purchasing behaviour.

The engineered features significantly enhanced the analytical capability of the dataset, enabling deeper customer segmentation and supporting the project's key business objectives. These features served as the foundation for subsequent SQL analysis, Power BI dashboard development, customer profiling, and the formulation of actionable business recommendations, including the promotional sunset strategy and the Ideal Customer Profile.

Following feature engineering, the dataset expanded from **17 cleaned attributes to 27 analytical features**, providing a comprehensive view of each customer's behavioural and commercial profile. The final engineered dataset contains the following attributes:

- Customer ID
- Age
- Gender
- Item Purchased
- Category
- Purchase Amount (USD)
- Location
- Size
- Color
- Season
- Review Rating
- Subscription Status
- Shipping Type
- Discount Applied
- Previous Purchases
- Payment Method
- Frequency of Purchases
- promo\_persona
- purchase\_frequency\_score
- tenure\_tier
- promo\_dependency\_score

- value\_tier
- satisfaction\_flag
- age\_group
- is\_premium\_shipping
- is\_digital\_payer
- loyalty\_score

With the analytical dataset prepared, the next phase of the project focuses on leveraging SQL to answer the core business questions related to customer value, loyalty, purchasing behaviour, geographic opportunities, and promotional strategy.

## 7. SQL-Driven Analysis

```
CREATE DATABASE CustomerValue;
GO
```

```
use CustomerValue
```

```
EXEC sp_rename 'dbo.engineered_dataset', 'customers';
```

```
--Section A — Customer Value Analysis
```

```
--Query A1: Customer Pyramid
```

```
--Business Question: What separates high-value customers from low-value customers?
```

```
SELECT
    value_tier,
    COUNT(*) AS customers,
    ROUND(AVG(Purchase_Amount_USD),2) AS avg_spend,
    ROUND(AVG(previous_purchases),1) AS avg_previous_purchases,
    ROUND(AVG(loyalty_score),3) AS avg_loyalty
FROM customers
GROUP BY value_tier
ORDER BY
CASE value_tier
    WHEN 'Premium' THEN 1
    WHEN 'High' THEN 2
    WHEN 'Medium' THEN 3
    WHEN 'Low' THEN 4
END;
```

**Output:**

| value_tier | customers | avg_spend | avg_previous_purchases | avg_loyalty |
|------------|-----------|-----------|------------------------|-------------|
| Premium    | 976       | 69        | 34                     | 0.433       |
| High       | 974       | 59        | 27                     | 0.128       |
| Medium     | 974       | 56        | 23                     | 0.032       |
| Low        | 976       | 53        | 15                     | 0.003       |

The customer pyramid analysis reveals a clear progression in customer value across the four value tiers. Premium customers exhibit the highest average purchase amount (\$69), the greatest purchase history (34 previous purchases), and the strongest loyalty score (0.433). In contrast, Low-value customers average only \$53 per purchase, have approximately 15 previous purchases, and exhibit almost negligible loyalty. These findings indicate that repeat purchasing behaviour, supported by sustained customer engagement, is a stronger determinant of long-term customer value than purchase amount alone.

This query validates the customer segmentation created during feature engineering. Premium customers are not only spending more per purchase but also returning much more frequently, which results in significantly higher loyalty scores. This suggests that long-term customer value is driven primarily by customer retention and repeat purchasing behaviour rather than individual transaction size. Therefore, retention strategies should focus on encouraging repeat engagement instead of only increasing basket value.

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## Query A2: Customer Composition Across Value Tiers

**Business Question:** *Are Premium customers primarily organic buyers or discount-driven customers?*

```

SELECT
  value_tier,
  promo_persona,
  COUNT(*) AS customers,
  ROUND(AVG(loyalty_score),3) AS avg_loyalty,
  ROUND(AVG(promo_dependency_score),3) AS avg_dependency
FROM customers
GROUP BY value_tier, promo_persona
ORDER BY
  CASE value_tier
    WHEN 'Premium' THEN 1
    WHEN 'High' THEN 2

```

```

    WHEN 'Medium' THEN 3
    WHEN 'Low' THEN 4
END,
CASE promo_persona
    WHEN 'Organic' THEN 1
    WHEN 'Subscriber' THEN 2
    WHEN 'Promo_Opportunist' THEN 3
END;

```

| value_tier | promo_persona     | customers | avg_loyalty | avg_dependency |
|------------|-------------------|-----------|-------------|----------------|
| Premium    | Organic           | 529       | 0.491       | 0              |
| Premium    | Subscriber        | 293       | 0.367       | 0.323          |
| Premium    | Promo_Opportunist | 154       | 0.362       | 0.324          |
| High       | Organic           | 577       | 0.158       | 0              |
| High       | Subscriber        | 257       | 0.089       | 0.533          |
| High       | Promo_Opportunist | 140       | 0.08        | 0.557          |
| Medium     | Organic           | 552       | 0.046       | 0              |
| Medium     | Subscriber        | 251       | 0.015       | 0.686          |
| Medium     | Promo_Opportunist | 171       | 0.016       | 0.676          |
| Low        | Organic           | 565       | 0.005       | 0              |
| Low        | Subscriber        | 252       | 0.001       | 0.812          |
| Low        | Promo_Opportunist | 159       | 0.001       | 0.815          |

This query examines the distribution of different customer personas within each value tier. By combining the engineered **Value Tier** and **Promotion Persona** features, it identifies whether high-value customers purchase because of genuine brand loyalty or because of promotional incentives. In addition, the query compares the average loyalty score and promotion dependency score across customer segments.

### Key Findings

- The **Premium** value tier consists of **529 Organic customers**, **293 Subscribers**, and **154 Promo Opportunists**. Organic customers account for approximately **54%** of all

Premium customers, making them the largest customer group within the highest-value segment.

- Organic Premium customers exhibit the **highest average loyalty score (0.491)** while maintaining **zero promotion dependency**, indicating that their purchasing behaviour is driven primarily by brand preference rather than promotional incentives.
- Although Subscribers and Promo Opportunists also appear within the Premium tier, both groups display **lower loyalty scores (0.367 and 0.362 respectively)** along with **moderate promotion dependency ( $\approx 0.323$ )**. **This suggests that promotional incentives continue to influence their purchasing behaviour.**
- A similar trend is observed across the High, Medium, and Low value tiers, where Organic customers consistently demonstrate higher loyalty than discount-driven customer groups.

### Business Interpretation

The analysis indicates that the brand's **most valuable customers are predominantly Organic buyers** who continue purchasing without relying on discounts. In contrast, customers acquired or retained primarily through promotions exhibit greater dependence on discounts and comparatively lower loyalty.

These findings suggest that promotional campaigns should be used selectively rather than uniformly across the customer base. **Organic Premium customers should be retained through personalized experiences and loyalty benefits instead of price reductions, while promotional spending should be focused on customer segments that genuinely require incentives to remain engaged.**

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### Query B1: Loyalty Analysis by Customer Persona

**Business Question:** *Who are genuinely loyal customers versus those who primarily purchase when discounts are available?*

SELECT

promo\_persona,

COUNT(\*) AS customers,

ROUND(AVG(loyalty\_score),3) AS avg\_loyalty,

ROUND(AVG(previous\_purchases),1) AS avg\_tenure,

ROUND(AVG(purchase\_frequency\_score),1) AS avg\_frequency

FROM customers

GROUP BY promo\_persona

ORDER BY avg\_loyalty DESC;

| promo_persona     | customers | avg_loyalty | avg_tenure | avg_frequency |
|-------------------|-----------|-------------|------------|---------------|
| Organic           | 2223      | 0.170       | 25         | 17            |
| Subscriber        | 1053      | 0.128       | 26         | 17            |
| Promo_Opportunist | 624       | 0.113       | 25         | 16            |

This query compares the three customer personas created during feature engineering—**Organic**, **Subscriber**, and **Promo Opportunist**—using four key metrics: customer count, average loyalty score, average previous purchases (tenure), and average purchase frequency. The objective is to determine whether promotional incentives are associated with stronger customer loyalty or whether organically acquired customers demonstrate superior long-term engagement.

### Key Findings

- The customer base consists of **2,223 Organic customers**, **1,053 Subscribers**, and **624 Promo Opportunists**, making Organic customers the largest customer segment.
- Organic customers exhibit the **highest average loyalty score (0.170)**, outperforming both Subscribers (0.128) and Promo Opportunists (0.113).
- The average customer tenure remains relatively similar across all three groups, ranging between **25 and 26 previous purchases**, indicating that purchase history alone does not explain differences in loyalty.
- Similarly, the average purchase frequency is nearly identical across the customer personas, with values between **16 and 17 annual purchase equivalents**.

### Business Interpretation

Although all three customer groups demonstrate comparable purchasing frequency and customer tenure, **Organic customers consistently achieve the highest loyalty scores. This suggests that genuine customer loyalty is influenced not only by purchasing behaviour but also by the absence of promotional dependence.**

The relatively lower loyalty scores observed among Subscribers and Promo Opportunists indicate that promotional incentives alone are insufficient to build strong long-term customer relationships. Instead, customers who continue purchasing without discounts appear to have

developed stronger brand affinity and are therefore more valuable from a long-term retention perspective.

These findings support the recommendation that retention strategies should prioritize strengthening relationships with Organic customers while gradually reducing reliance on discount-based retention programs for customers exhibiting higher promotional dependence.

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## Query B2: Customer Segmentation by Promotion Dependency

**Business Question:** *Which customers are genuinely loyal and which are primarily driven by promotional incentives?*

WITH DependencyBands AS

```
(  
  SELECT  
    CASE  
      WHEN promo_dependency_score = 0 THEN 'Organic'  
      WHEN promo_dependency_score < 0.35 THEN 'Low Dependency'  
      WHEN promo_dependency_score < 0.65 THEN 'Moderate Dependency'  
      ELSE 'High Dependency'  
    END AS dependency_band,  
    loyalty_score  
  FROM customers  
)
```

```
SELECT  
  dependency_band,  
  COUNT(*) AS customers,  
  ROUND(AVG(loyalty_score),3) AS avg_loyalty  
FROM DependencyBands  
GROUP BY dependency_band  
ORDER BY  
CASE dependency_band
```

```
WHEN 'Organic' THEN 1

WHEN 'Low Dependency' THEN 2

WHEN 'Moderate Dependency' THEN 3

WHEN 'High Dependency' THEN 4

END;
```

| dependency_band     | customers | avg_loyalty |
|---------------------|-----------|-------------|
| Organic             | 2230      | 0.173       |
| Low Dependency      | 264       | 0.503       |
| Moderate Dependency | 705       | 0.084       |
| High Dependency     | 701       | 0.007       |

This query classifies customers into four promotion dependency bands based on the engineered Promotion Dependency Score. The objective is to evaluate how customer loyalty varies with increasing reliance on promotional discounts.

### Key Findings

- The **Organic** segment represents the largest customer group with **2,230 customers**, exhibiting an average loyalty score of **0.173**.
- The **Low Dependency** segment contains only **264 customers** but achieves the **highest average loyalty score (0.503)** among all customer groups.
- Customers with **Moderate Dependency** demonstrate considerably lower loyalty (0.084), while those with **High Dependency** exhibit an almost negligible average loyalty score (0.007).

### Business Interpretation

The results indicate that occasional promotional usage does not necessarily imply poor customer loyalty. Customers within the **Low Dependency** segment continue to purchase frequently and maintain long-term relationships with the brand despite receiving limited



promotional support, making them the strongest candidates for gradually reducing discount offers.

In contrast, customers classified as **High Dependency** rely heavily on promotions and demonstrate extremely low loyalty. These customers are more likely to discontinue purchasing if discounts are withdrawn abruptly and therefore require a carefully managed promotional strategy.

The Organic segment includes customers across all value and engagement levels, resulting in a moderate average loyalty score despite having no promotional dependence. Consequently, promotion dependency should be evaluated alongside purchasing behaviour rather than in isolation when designing customer retention strategies.

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## Query C1: Frequency–Tenure Matrix

**Business Question:** *What behavioral patterns predict high customer value over time?*

SELECT

tenure\_tier,

frequency\_of\_purchases,

COUNT(\*) AS customers,

ROUND(AVG(loyalty\_score),3) AS avg\_loyalty,

ROUND(AVG(previous\_purchases),1) AS avg\_tenure,

ROUND(

100.0 \*

SUM(CASE

WHEN value\_tier = 'Premium' THEN 1

ELSE 0

END)

/ COUNT(\*),

1

) AS premium\_pct

FROM customers

GROUP BY

tenure\_tier,  
  
frequency\_of\_purchases

ORDER BY

premium\_pct DESC,

CASE tenure\_tier

WHEN 'Loyal' THEN 1

WHEN 'Established' THEN 2

WHEN 'Growing' THEN 3

WHEN 'New' THEN 4

END,

CASE frequency\_of\_purchases

WHEN 'Weekly' THEN 1

WHEN 'Bi-Weekly' THEN 2

WHEN 'Monthly' THEN 3

WHEN 'Quarterly' THEN 4

WHEN 'Annually' THEN 5

END;

| tenure_tier | frequency_of_purchases | customers | avg_loyalty | avg_tenure | premium_pct      |
|-------------|------------------------|-----------|-------------|------------|------------------|
| Loyal       | Weekly                 | 128       | 0.949       | 44         | 100.000000000000 |
| Established | Weekly                 | 151       | 0.632       | 31         | 97.400000000000  |
| Loyal       | Bi-Weekly              | 249       | 0.407       | 44         | 85.900000000000  |
| Growing     | Weekly                 | 129       | 0.356       | 19         | 74.400000000000  |

|             |           |     |       |    |                   |
|-------------|-----------|-----|-------|----|-------------------|
| Established | Bi-Weekly | 284 | 0.270 | 31 | 73.20000000000000 |
| Loyal       | Monthly   | 118 | 0.168 | 44 | 39.80000000000000 |
| Growing     | Bi-Weekly | 259 | 0.150 | 19 | 37.80000000000000 |
| New         | Weekly    | 131 | 0.097 | 6  | 16.00000000000000 |
| Established | Monthly   | 156 | 0.116 | 32 | 10.90000000000000 |
| Loyal       | Quarterly | 309 | 0.044 | 44 | 0.00000000000000  |
| Loyal       | Annually  | 124 | 0     | 44 | 0.00000000000000  |
| Established | Quarterly | 274 | 0.029 | 32 | 0.00000000000000  |
| Established | Annually  | 142 | 0     | 31 | 0.00000000000000  |
| Growing     | Monthly   | 145 | 0.060 | 19 | 0.00000000000000  |
| Growing     | Quarterly | 268 | 0.016 | 19 | 0.00000000000000  |
| Growing     | Annually  | 150 | 0     | 19 | 0.00000000000000  |
| New         | Bi-Weekly | 297 | 0.049 | 7  | 0.00000000000000  |

|     |           |     |       |   |                |
|-----|-----------|-----|-------|---|----------------|
| New | Monthly   | 134 | 0.017 | 6 | 0.000000000000 |
| New | Quarterly | 296 | 0.005 | 6 | 0.000000000000 |
| New | Annually  | 156 | 0     | 7 | 0.000000000000 |

This query investigates the combined influence of **customer tenure** and **purchase frequency** on long-term customer value. Customers are grouped according to their tenure tier and purchase frequency, after which key performance indicators such as average loyalty score, average previous purchases, and the percentage of Premium customers are calculated. The objective is to identify the behavioral characteristics that consistently distinguish high-value customers from the rest of the customer base.

**Key Findings**

- Customers belonging to the **Loyal** tenure tier and purchasing **Weekly** demonstrated the strongest performance, with **100% of customers classified as Premium** and the highest average loyalty score of **0.949**.
- Customers in the **Established** tenure tier who purchased **Weekly** also showed exceptional performance, with **97.4% Premium customers** and an average loyalty score of **0.632**.
- Customers purchasing **Bi-Weekly** also exhibited strong performance, particularly within the Loyal and Established tenure groups, where approximately **86%** and **73%** of customers respectively belonged to the Premium value tier.
- As purchase frequency decreased from **Weekly** to **Monthly**, **Quarterly**, and **Annually**, both the average loyalty score and the proportion of Premium customers declined sharply. Customers purchasing Quarterly or Annually contributed virtually no Premium customers regardless of their tenure level.
- While longer customer tenure positively influenced loyalty, the results indicate that **high purchase frequency combined with long tenure** is the strongest behavioral predictor of long-term customer value.

**Business Interpretation**

The analysis demonstrates that customer value is driven by a combination of **purchase frequency** and **relationship duration**, rather than either factor individually. Customers who engage with the brand frequently over an extended period consistently generate the highest loyalty scores and are significantly more likely to belong to the Premium value tier.

These findings suggest that customer retention strategies should focus on increasing purchase frequency among newer customers while simultaneously encouraging long-term engagement.

Marketing initiatives such as personalized recommendations, subscription programs, and loyalty rewards should be designed to transition customers from infrequent purchasing behavior toward regular purchasing patterns, ultimately increasing the proportion of Premium customers within the customer base.

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**Query C2: Relationship Between Customer Satisfaction and Loyalty**

**Business Question:** *Does customer satisfaction influence long-term customer loyalty?*

```
SELECT
    satisfaction_flag,
    ROUND(AVG(loyalty_score),3) AS avg_loyalty,
    ROUND(AVG(previous_purchases),1) AS avg_tenure
FROM customers
GROUP BY satisfaction_flag;
```

| satisfaction_flag | avg_loyalty | avg_tenure |
|-------------------|-------------|------------|
| 0                 | 0.135       | 25         |
| 1                 | 0.171       | 25         |

This query compares customers based on the engineered **Satisfaction Flag**, which classifies customers according to their review ratings. For each group, the average loyalty score and average previous purchases are calculated to evaluate whether customer satisfaction is associated with stronger long-term engagement.

**Key Findings**

- Customers classified as **Satisfied** achieved an average loyalty score of **0.171**, compared to **0.135** for customers who were not classified as satisfied.
- The average number of previous purchases remained nearly identical for both groups, with each averaging approximately **25 previous purchases**.

- The satisfied customer segment therefore demonstrates a noticeably higher level of loyalty despite having a similar purchase history.

### Business Interpretation

The analysis suggests that customer satisfaction contributes positively to long-term loyalty. Since both customer groups exhibit comparable purchase histories, the higher loyalty observed among satisfied customers cannot be attributed solely to customer tenure. Instead, it indicates that maintaining a positive customer experience strengthens customer commitment beyond simply encouraging repeat purchases.

These findings highlight the importance of investing in product quality, customer service, and post-purchase experience. Improving customer satisfaction can enhance customer loyalty and retention without necessarily increasing promotional spending.

---

### Query D1: Category-wise Customer Retention Analysis

**Business Question:** *Which product categories are associated with stronger customer retention and long-term loyalty?*

SELECT

category,

ROUND(AVG(previous\_purchases),1)

AS avg\_previous\_purchases,

ROUND(AVG(loyalty\_score),3)

AS avg\_loyalty

FROM customers

GROUP BY category

ORDER BY avg\_previous\_purchases DESC;

| category | avg_previous_purchases | avg_loyalty |
|----------|------------------------|-------------|
|----------|------------------------|-------------|

|             |    |       |
|-------------|----|-------|
| Clothing    | 25 | 0.146 |
| Accessories | 25 | 0.151 |
| Footwear    | 25 | 0.156 |
| Outerwear   | 24 | 0.154 |

This query evaluates customer retention across different product categories by calculating the average number of previous purchases and the average loyalty score for customers purchasing within each category. The objective is to determine whether certain product categories naturally encourage stronger long-term customer relationships.

#### Key Findings

- Customers across all product categories exhibited remarkably similar purchasing histories, with an average of approximately **24–25 previous purchases**, indicating that no category demonstrated significantly higher customer retention than the others.
- **Footwear** recorded the highest average loyalty score (**0.156**), followed closely by **Outerwear (0.154)** and **Accessories (0.151)**.
- **Clothing** customers displayed the lowest average loyalty score (**0.146**), although the difference from the other categories was relatively small.

#### Business Interpretation

The analysis indicates that product category alone is not a major determinant of long-term customer retention. Customers purchasing different categories demonstrate similar purchasing histories, while only modest variations are observed in loyalty levels.

These findings suggest that customer value is influenced more strongly by behavioural factors such as purchase frequency, promotional dependence, and customer engagement than by the product category itself. Consequently, retention strategies should focus primarily on customer behaviour rather than allocating resources solely based on product category.

---

### Query D2: Seasonal Customer Behaviour Analysis

**Business Question:** *Which seasons are associated with higher customer retention and long-term loyalty?*

```

SELECT

    season,

    ROUND(AVG(previous_purchases),1)

    AS avg_previous_purchases,

    ROUND(AVG(loyalty_score),3)

    AS avg_loyalty

FROM customers

GROUP BY season

ORDER BY avg_previous_purchases DESC;

```

| season | avg_previous_purchases | avg_loyalty |
|--------|------------------------|-------------|
| Summer | 25                     | 0.147       |
| Spring | 25                     | 0.148       |
| Winter | 25                     | 0.161       |
| Fall   | 24                     | 0.144       |

This query evaluates customer behaviour across different seasons by calculating the average number of previous purchases and the average loyalty score for customers purchasing during each season. The objective is to determine whether seasonal purchasing patterns are associated with stronger customer retention.

### Key Findings

- Customers purchasing across all four seasons demonstrated very similar purchasing histories, averaging approximately **24–25 previous purchases**, indicating that customer tenure remains relatively consistent throughout the year.
- Winter** customers exhibited the highest average loyalty score (**0.161**), suggesting slightly stronger long-term customer engagement compared to other seasons.



- **Fall** recorded the lowest average loyalty score (**0.144**) and marginally lower average previous purchases, indicating comparatively weaker customer engagement.
- The differences between seasons were relatively small, suggesting that seasonal purchasing behaviour has only a limited influence on long-term customer value.

### Business Interpretation

The analysis indicates that seasonality is not a major determinant of customer retention or loyalty. While Winter customers demonstrate marginally stronger loyalty, customer value appears to be influenced far more by behavioural characteristics such as purchase frequency, promotional dependence, and customer tenure than by seasonal purchasing patterns.

These findings suggest that seasonal marketing campaigns should be used primarily to support customer engagement rather than as the primary mechanism for improving customer retention. Greater emphasis should instead be placed on behavioural segmentation and personalized retention strategies.

### Query E1: Geographic Opportunity Analysis

**Business Question:** *Which geographic markets demonstrate strong organic demand and which rely primarily on promotional incentives?*

```
SELECT
```

```
    location,
```

```
    COUNT(*) AS customers,
```

```
    ROUND(
```

```
        100.0 *
```

```
        SUM(
```

```
            CASE
```

```
                WHEN promo_persona='Organic'
```

```
            THEN 1
```

```
            ELSE 0
```

```
        END
```

```
    ) / COUNT(*),1
```

```
    ) AS organic_pct,
```

```

ROUND(AVG(loyalty_score),3)

AS avg_loyalty,

ROUND(AVG(promo_dependency_score),3)

AS avg_dependency

FROM customers

GROUP BY location

ORDER BY avg_loyalty DESC;

```

| location       | customers | organic_pct      | avg_loyalty | avg_dependency |
|----------------|-----------|------------------|-------------|----------------|
| Hawaii         | 65        | 50.8000000000000 | 0.216       | 0.249          |
| Tennessee      | 77        | 63.6000000000000 | 0.203       | 0.215          |
| Alaska         | 72        | 59.7000000000000 | 0.195       | 0.221          |
| Alabama        | 89        | 59.6000000000000 | 0.192       | 0.247          |
| Wyoming        | 71        | 57.7000000000000 | 0.188       | 0.228          |
| Georgia        | 79        | 59.5000000000000 | 0.187       | 0.228          |
| Pennsylvania   | 74        | 55.4000000000000 | 0.185       | 0.257          |
| North Carolina | 78        | 55.1000000000000 | 0.184       | 0.252          |
| New York       | 87        | 58.6000000000000 | 0.183       | 0.257          |
| Arizona        | 65        | 66.2000000000000 | 0.182       | 0.190          |
| Oregon         | 74        | 48.6000000000000 | 0.180       | 0.290          |

|              |    |                 |       |       |
|--------------|----|-----------------|-------|-------|
| Illinois     | 92 | 59.800000000000 | 0.172 | 0.212 |
| Minnesota    | 88 | 55.700000000000 | 0.171 | 0.264 |
| Maryland     | 86 | 55.800000000000 | 0.170 | 0.261 |
| Kentucky     | 79 | 57.000000000000 | 0.164 | 0.253 |
| Louisiana    | 84 | 58.300000000000 | 0.163 | 0.218 |
| Indiana      | 79 | 43.000000000000 | 0.162 | 0.329 |
| Iowa         | 69 | 47.800000000000 | 0.162 | 0.279 |
| Kansas       | 63 | 76.200000000000 | 0.158 | 0.136 |
| Virginia     | 77 | 62.300000000000 | 0.158 | 0.225 |
| Rhode Island | 63 | 60.300000000000 | 0.157 | 0.237 |
| Ohio         | 77 | 55.800000000000 | 0.155 | 0.242 |
| Oklahoma     | 75 | 52.000000000000 | 0.153 | 0.291 |
| Nevada       | 87 | 52.900000000000 | 0.152 | 0.279 |
| Florida      | 68 | 54.400000000000 | 0.149 | 0.261 |
| Mississippi  | 80 | 52.500000000000 | 0.148 | 0.285 |
| South Dakota | 70 | 62.900000000000 | 0.148 | 0.221 |
| New Mexico   | 81 | 55.600000000000 | 0.148 | 0.242 |

|               |    |                 |       |       |
|---------------|----|-----------------|-------|-------|
| Colorado      | 75 | 58.700000000000 | 0.146 | 0.243 |
| Massachusetts | 72 | 51.400000000000 | 0.143 | 0.280 |
| New Jersey    | 67 | 59.700000000000 | 0.142 | 0.256 |
| California    | 95 | 57.900000000000 | 0.141 | 0.269 |
| Maine         | 77 | 64.900000000000 | 0.133 | 0.209 |
| Arkansas      | 79 | 53.200000000000 | 0.133 | 0.269 |
| Montana       | 96 | 62.500000000000 | 0.132 | 0.206 |
| West Virginia | 81 | 50.600000000000 | 0.130 | 0.314 |
| Washington    | 73 | 56.200000000000 | 0.130 | 0.257 |
| Missouri      | 81 | 50.600000000000 | 0.129 | 0.285 |
| Texas         | 77 | 63.600000000000 | 0.123 | 0.215 |
| Nebraska      | 87 | 57.500000000000 | 0.123 | 0.259 |
| North Dakota  | 83 | 54.200000000000 | 0.122 | 0.270 |
| Utah          | 71 | 53.500000000000 | 0.120 | 0.276 |
| Vermont       | 85 | 61.200000000000 | 0.119 | 0.229 |
| New Hampshire | 71 | 56.300000000000 | 0.114 | 0.259 |
| Connecticut   | 78 | 66.700000000000 | 0.110 | 0.184 |

|                |    |                 |       |       |
|----------------|----|-----------------|-------|-------|
| Idaho          | 93 | 59.100000000000 | 0.109 | 0.248 |
| Michigan       | 73 | 60.300000000000 | 0.109 | 0.234 |
| Wisconsin      | 75 | 52.000000000000 | 0.109 | 0.319 |
| Delaware       | 86 | 54.700000000000 | 0.107 | 0.283 |
| South Carolina | 76 | 50.000000000000 | 0.106 | 0.321 |

This query evaluates customer behaviour across different geographic locations by measuring the proportion of Organic customers, the average loyalty score, and the average promotion dependency score. The objective is to identify markets where customer demand is driven primarily by brand affinity rather than promotional discounts.

### Key Findings

- **Hawaii** recorded the highest average loyalty score (**0.216**), followed by **Tennessee (0.203)** and **Alaska (0.195)**, indicating stronger long-term customer engagement in these markets.
- Several states demonstrated a high proportion of **Organic customers**, including **Kansas (76.2%)**, **Connecticut (66.7%)**, **Arizona (66.2%)**, **Maine (64.9%)**, and **Tennessee (63.6%)**. These markets appear to generate stronger demand without heavy dependence on promotional discounts.
- States such as **Indiana (43.0%)**, **Oregon (48.6%)**, and **South Carolina (50.0%)** recorded comparatively lower percentages of Organic customers together with higher promotion dependency scores, suggesting greater reliance on promotional incentives to generate sales.
- Promotion dependency varied across locations, with **Arizona (0.190)**, **Kansas (0.136)**, and **Tennessee (0.215)** displaying relatively low dependency, whereas **Indiana (0.329)**, **South Carolina (0.321)**, and **Wisconsin (0.319)** exhibited substantially higher dependence on promotional offers.

### Business Interpretation

The analysis reveals significant geographic variation in customer behaviour. Markets such as **Tennessee**, **Arizona**, and **Kansas** combine relatively high loyalty with a large proportion of Organic customers and comparatively low promotion dependency, making them attractive candidates for premium positioning and reduced promotional spending.

Conversely, markets exhibiting higher promotion dependency and lower loyalty should continue receiving targeted promotional support while additional efforts are made to improve long-term customer engagement. These findings demonstrate that promotional strategy should be geographically differentiated rather than applied uniformly across all markets.

---

**Query E2: High-Potential Markets for Strategic Investment**

**Business Question:** *Which geographic markets deserve greater investment based on customer spending and loyalty?*

```
SELECT
    location,
    COUNT(*) AS customers,
    ROUND(AVG(Purchase_Amount_USD),2) AS avg_spend,
    ROUND(AVG(loyalty_score),3) AS avg_loyalty
FROM customers
GROUP BY location
HAVING
    AVG(Purchase_Amount_USD) > 60
    AND AVG(loyalty_score) > 0.15
ORDER BY
    avg_spend DESC;
```

| location     | customers | avg_spend | avg_loyalty |
|--------------|-----------|-----------|-------------|
| Alaska       | 72        | 67        | 0.195       |
| Pennsylvania | 74        | 66        | 0.185       |
| Arizona      | 65        | 66        | 0.182       |
| Nevada       | 87        | 63        | 0.152       |

|              |    |    |       |
|--------------|----|----|-------|
| Virginia     | 77 | 62 | 0.158 |
| Tennessee    | 77 | 61 | 0.203 |
| Rhode Island | 63 | 61 | 0.157 |
| Illinois     | 92 | 61 | 0.172 |

This query identifies geographic markets that satisfy two business criteria: an average purchase amount greater than \$60 and an average loyalty score exceeding 0.15. These thresholds were used to identify markets that generate both relatively high revenue and stronger customer relationships, making them attractive candidates for future business expansion.

#### Key Findings

- Eight geographic markets satisfied the predefined selection criteria, namely **Alaska, Pennsylvania, Arizona, Nevada, Virginia, Tennessee, Rhode Island, and Illinois**.
- **Alaska** recorded the highest average customer spending (**\$67**) while also maintaining a strong average loyalty score (**0.195**), indicating excellent commercial potential.
- **Tennessee** achieved the highest average loyalty score (**0.203**) among all selected markets while maintaining above-average customer spending, suggesting a highly engaged customer base.
- **Pennsylvania** and **Arizona** also demonstrated a favorable combination of high customer spending and strong loyalty, making them attractive markets for future marketing and customer acquisition initiatives.

#### Business Interpretation

The identified markets demonstrate a balance between customer spending and customer loyalty, making them suitable candidates for increased marketing investment and business expansion. Rather than distributing marketing resources uniformly across all regions, the company can prioritize these high-potential markets to maximize customer lifetime value and improve the return on marketing expenditure.

These markets may benefit from targeted acquisition campaigns, personalized engagement strategies, and premium product positioning to further strengthen customer relationships while supporting sustainable revenue growth.

---

## Query F1: Ideal Customer Profile

**Business Question:** *What does the brand's ideal customer profile look like?*

SELECT

age\_group,

gender,

category,

COUNT(\*) AS customers,

ROUND(AVG(loyalty\_score),3)

AS avg\_loyalty

FROM customers

WHERE

value\_tier='Premium'

AND promo\_persona='Organic'

GROUP BY

age\_group,

gender,

category

ORDER BY avg\_loyalty DESC;

| age_group | gender | category | customers | avg_loyalty |
|-----------|--------|----------|-----------|-------------|
|-----------|--------|----------|-----------|-------------|



|       |        |             |    |       |
|-------|--------|-------------|----|-------|
| 36-50 | Female | Accessories | 20 | 0.588 |
| 36-50 | Male   | Footwear    | 11 | 0.572 |
| 51-70 | Female | Outerwear   | 8  | 0.557 |
| 51-70 | Female | Footwear    | 21 | 0.556 |
| 51-70 | Male   | Clothing    | 45 | 0.555 |
| 26-35 | Male   | Outerwear   | 4  | 0.541 |
| 26-35 | Female | Footwear    | 10 | 0.538 |
| 51-70 | Male   | Accessories | 31 | 0.526 |
| 26-35 | Male   | Accessories | 15 | 0.522 |
| 18-25 | Female | Accessories | 15 | 0.513 |
| 51-70 | Male   | Outerwear   | 5  | 0.512 |
| 51-70 | Female | Accessories | 29 | 0.511 |
| 51-70 | Male   | Footwear    | 14 | 0.509 |

|       |        |             |    |       |
|-------|--------|-------------|----|-------|
| 26-35 | Male   | Clothing    | 20 | 0.508 |
| 36-50 | Female | Outerwear   | 3  | 0.507 |
| 26-35 | Female | Clothing    | 33 | 0.502 |
| 36-50 | Male   | Clothing    | 27 | 0.499 |
| 36-50 | Male   | Outerwear   | 6  | 0.498 |
| 51-70 | Female | Clothing    | 50 | 0.476 |
| 18-25 | Male   | Outerwear   | 7  | 0.472 |
| 18-25 | Female | Outerwear   | 6  | 0.468 |
| 36-50 | Male   | Accessories | 25 | 0.451 |
| 18-25 | Female | Clothing    | 13 | 0.440 |
| 26-35 | Female | Accessories | 13 | 0.430 |
| 18-25 | Male   | Footwear    | 5  | 0.427 |
| 18-25 | Male   | Clothing    | 19 | 0.421 |

|       |        |             |    |       |
|-------|--------|-------------|----|-------|
| 36-50 | Female | Clothing    | 36 | 0.411 |
| 36-50 | Female | Footwear    | 12 | 0.401 |
| 18-25 | Female | Footwear    | 6  | 0.400 |
| 18-25 | Male   | Accessories | 11 | 0.379 |
| 26-35 | Female | Outerwear   | 4  | 0.349 |
| 26-35 | Male   | Footwear    | 5  | 0.346 |

This query identifies the characteristics of the brand's most valuable customers by analyzing only those customers who belong to the **Premium Value Tier** and are classified as **Organic** buyers. These customers represent the highest-value segment because they generate substantial customer value while purchasing without dependence on promotional discounts.

### Key Findings

- The highest loyalty scores were primarily observed among customers belonging to the **36–50** and **51–70** age groups, indicating stronger long-term engagement among mature customers.
- **Accessories**, **Footwear**, and **Clothing** consistently appeared among the highest-performing product categories within the Premium Organic customer segment.
- Although a few smaller customer groups achieved exceptionally high loyalty scores, the largest concentrations of Premium Organic customers were found within the Clothing and Accessories categories, suggesting greater commercial significance.
- Across the identified segments, customers exhibited strong purchasing behaviour while maintaining minimal promotional dependence, confirming that genuine customer value is driven by sustained engagement rather than discount-driven purchasing.

### Business Interpretation

The analysis suggests that the brand's ideal customer is typically a **Premium Organic customer** belonging to the **36–50 or 51–70 age groups**, purchasing primarily within the Clothing, Accessories, or Footwear categories. These customers demonstrate consistently high loyalty while requiring little promotional support, making them the most profitable customer segment for long-term business growth.

Marketing efforts should therefore prioritize acquiring similar customer profiles through targeted advertising, personalized engagement, and loyalty-focused retention strategies rather than relying heavily on discount-based customer acquisition.

### Query G1: Promotion Sunset Strategy

**Business Question:** *Which customer segments are sufficiently loyal to gradually reduce promotional discounts without significantly affecting customer retention?*

```
SELECT
    tenure_tier,
    frequency_of_purchases,

    COUNT(*) AS customers,

    ROUND(AVG(loyalty_score),3)

    AS avg_loyalty,

    ROUND(AVG(promo_dependency_score),3)

    AS avg_dependency
FROM customers
WHERE promo_persona='Subscriber'
GROUP BY
    tenure_tier,
    frequency_of_purchases
ORDER BY avg_loyalty DESC;
```

|             |                        |           |             |                |
|-------------|------------------------|-----------|-------------|----------------|
| tenure_tier | frequency_of_purchases | customers | avg_loyalty | avg_dependency |
|-------------|------------------------|-----------|-------------|----------------|

|             |           |    |       |       |
|-------------|-----------|----|-------|-------|
| Loyal       | Weekly    | 38 | 0.945 | 0.050 |
| Established | Weekly    | 46 | 0.569 | 0.183 |
| Loyal       | Bi-Weekly | 70 | 0.320 | 0.314 |
| Growing     | Weekly    | 40 | 0.280 | 0.312 |
| Established | Bi-Weekly | 92 | 0.188 | 0.440 |
| Loyal       | Monthly   | 33 | 0.116 | 0.447 |
| Growing     | Bi-Weekly | 64 | 0.091 | 0.562 |
| New         | Weekly    | 33 | 0.075 | 0.441 |
| Established | Monthly   | 43 | 0.065 | 0.574 |
| Growing     | Monthly   | 37 | 0.029 | 0.695 |
| Loyal       | Quarterly | 76 | 0.027 | 0.525 |
| New         | Bi-Weekly | 67 | 0.022 | 0.693 |
| Established | Quarterly | 76 | 0.015 | 0.650 |
| New         | Monthly   | 36 | 0.005 | 0.832 |
| Growing     | Quarterly | 69 | 0.005 | 0.782 |
| New         | Quarterly | 73 | 0.001 | 0.910 |
| New         | Annually  | 43 | 0     | 0.927 |

|             |          |    |   |       |
|-------------|----------|----|---|-------|
| Growing     | Annually | 46 | 0 | 0.811 |
| Loyal       | Annually | 34 | 0 | 0.559 |
| Established | Annually | 37 | 0 | 0.683 |

This query evaluates customers belonging to the **Subscriber** segment by comparing their tenure, purchase frequency, loyalty score, and promotion dependency score. The objective is to identify customer groups that continue purchasing primarily because of brand loyalty rather than promotional incentives.

### Key Findings

- **Loyal customers purchasing Weekly** exhibited the highest loyalty score (**0.945**) while maintaining an extremely low promotion dependency (**0.050**). This segment represents the strongest candidate for a gradual reduction in promotional discounts.
- **Established Weekly** customers also demonstrated high loyalty (**0.569**) with relatively low promotion dependency (**0.183**), indicating that they are likely to continue purchasing even if promotional intensity is reduced.
- Customers purchasing **Bi-Weekly** within the Loyal and Established tenure groups maintained moderate loyalty while exhibiting acceptable dependency levels, making them suitable candidates for phased promotional optimization.
- In contrast, **New** and **Growing** customers purchasing Monthly, Quarterly, or Annually displayed very low loyalty scores together with high promotion dependency. These segments continue to rely heavily on promotional incentives and should therefore retain existing discount programs.

### Business Interpretation

The analysis demonstrates that promotional discounts should not be removed uniformly across the customer base. Instead, the organization should adopt a **phased promotional sunset strategy**, beginning with highly loyal, frequent purchasers who exhibit minimal dependence on discounts. Reducing promotional spending within these mature customer segments can improve profit margins while minimizing the risk of customer attrition.

Conversely, newer customers and those demonstrating high promotion dependency should continue receiving promotional support until stronger purchasing habits and customer loyalty are established.

### Business Recommendation

| Customer Segment                   | Recommendation                  | Reason                                     |
|------------------------------------|---------------------------------|--------------------------------------------|
| Loyal + Weekly                     | Phase out discounts first       | Extremely high loyalty, minimal dependency |
| Established + Weekly               | Reduce discounts gradually      | Strong loyalty, low dependency             |
| Loyal + Bi-Weekly                  | Pilot reduced promotions        | Moderate dependency, mature customers      |
| Growing + Weekly                   | Test partial discount reduction | Loyalty developing                         |
| New / Monthly / Quarterly / Annual | Continue discounts              | Highly dependent on promotions             |